

Course Name: ETHICAL HACKING

Assignment- Week 3

TYPE OF QUESTION: MCQ/MSQ/SA

Number of questions: 10

Total mark: 10 x 1 = 10

QUESTION 1:

Which of the following statement(s) is/are true regarding connectionless routing used by the IP protocol?

- a. Each packet is treated as an independent unit for routing decisions.
- b. A fixed path is established before any packet is transmitted.
- c. Different packets between the same source and destination may follow different paths.
- d. IP protocol uses connection-oriented routing.

Correct Answer: a, c

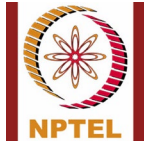
Detailed Solution: IP uses connectionless routing, where each packet is treated independently and may follow a different path even between the same source-destination pair. IP does not establish a fixed path beforehand (that is connection-oriented behavior).

Thus, the correct options are (a) and (c).

QUESTION 2:

Which of the following statement(s) is/are **false** regarding direct and indirect packet delivery?

- a. Direct delivery occurs when the destination host and deliverer are on the same physical network.
- b. Indirect delivery occurs when the destination host is not on the same network as the deliverer.
- c. In direct delivery, packets always pass through at least one router.
- d. In indirect delivery, the packet is routed hop by hop until it reaches a router on the destination's network.
- e. Direct delivery can occur between the last router and the destination host.



Correct Answer: c

Detailed Solution: Direct delivery occurs when source and destination share the same physical network, or between the last router and the destination host — hence it does NOT always require passing through a router, making (c) false. All other statements correctly describe direct/indirect delivery.

Thus, the correct (false) option is (c).

QUESTION 3:

Which routing table flag indicates that a route is specific to a single host rather than an entire network?

- a. U
- b. G
- c. H
- d. D
- e. M

Correct Answer: c

Detailed Solution: The H flag in a routing table entry indicates that the route is host-specific (a route to a single host), as opposed to a route to an entire network.

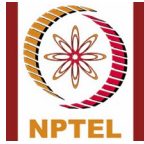
Thus the correct option is (c).

QUESTION 4:

In the OSPF routing protocol, which type of packet is used to determine whether a neighboring router is active?

- a. Link State Update
- b. Link State Request
- c. Hello packet
- d. Link State Acknowledgement
- e. TCP SYN packet

Correct Answer: c



Detailed Solution: OSPF uses "Hello" packets periodically exchanged between neighbouring routers to check whether a neighbour is alive/active. If Hello packets are not received within a timeout period, the neighbour is considered down.

Thus, the correct option is (c).

QUESTION 5:

Which type of address should be used when a packet needs to be delivered to every host within a specific network?

- a. Unicast address
- b. Broadcast address
- c. Anycast address
- d. None of these

Correct Answer: b

Detailed Solution: A broadcast address is used to deliver a packet to all hosts within a given network or subnetwork. Unicast targets a single specific host, and anycast targets the nearest of a group of hosts.

Thus, the correct option is (b).

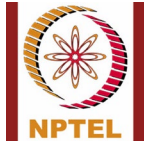
QUESTION 6:

Which of the following statement(s) is/are **false** regarding routing protocols?

- a. RIP uses distance vector routing and can suffer from the count-to-infinity problem.
- b. OSPF is a link-state routing protocol.
- c. BGP is used for routing within a single autonomous system.
- d. OSPF uses Dijkstra's algorithm to compute shortest paths.
- e. RIP is typically used for routing between different autonomous systems.

Correct Answer: c, e

Detailed Solution: BGP is an exterior routing protocol used for routing between different autonomous systems, not within a single AS, making (c) false. RIP is an interior protocol used within a single AS, not between different ASes, making (e) false. RIP's distance vector approach can indeed suffer from count-to-infinity, and OSPF is a link-state protocol using Dijkstra's algorithm - both correctly stated.



Thus, the correct (false) options are (c) and (e).

QUESTION 7:

What is the address used to represent the default route in a routing table (in IPv4)?

- a. 127.0.0.1
- b. 255.255.255.255
- c. 0.0.0.0
- d. 192.0.0.1
- e. None of these

Correct Answer: c

Detailed Solution: The default route, also called the "gateway of last resort," is represented by the address 0.0.0.0 (with subnet mask 0.0.0.0, or /0 in CIDR), and is used for packets that don't match any other entry in the routing table.

Thus, the correct option is (c).

QUESTION 8:

Which of the following statement(s) is/are **false** regarding IPv6?

- a. IPv6 supports address classes similar to IPv4 (Class A, B, C).
- b. IPv6 addresses are typically represented in hexadecimal, separated by colons.
- c. IPv6 eliminates the need for fragmentation by intermediate routers.
- d. The base header size of IPv6 is variable, ranging from 20 to 60 bytes.

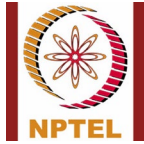
Correct Answer: a, d

Detailed Solution: IPv6 does not use classful addressing like IPv4 (no Class A/B/C), making (a) false. The IPv6 base header has a fixed size of 40 bytes (not variable 20-60 bytes, which describes IPv4), making (d) false. IPv6 addresses are indeed hexadecimal and colon-separated, fragmentation in IPv6 is done only by the source (not intermediate routers).

Thus, the correct (false) options are (a) and (d).

QUESTION 9:

In IPv6 base header _____ bits are used for Hop Limit?



Correct Answer: 8

Detailed Solution: The Hop Limit field in the IPv6 base header is 8 bits wide, serving a similar function to the TTL field in IPv4 - it is decremented at each hop, and the packet is discarded when it reaches 0.

Thus, the correct answer is 8.

QUESTION 10:

An organization is assigned 172.20.64.0/20 network block. _____ number of valid host addresses are available in this block.

Correct Answer: 4094

Detail Solution: A /20 prefix leaves 12 host bits, giving $2^{12} = 4096$ addresses. Subtracting the network address and broadcast address leaves 4094 usable host addresses.

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